SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA, INDORE

Shri Vaishnav Institute of Information Technology (SVIIT)

Department of Computer Science & Engineering

MAJOR PROJECT (BTCS706N) COURSEWORK REPORT

UML ARCHITECTURE & SYSTEM DESIGN DOSSIER

e-Portal for Case Management Hearing of Commercial Suits

(Governed by Order XV-A of Code of Civil Procedure, 1908 & Commercial Courts Act, 2015)

ACADEMIC SESSION:	**2026 – 2027 (Semester VII B.Tech CSE)**
TEAM IDENTIFIER:	**SVVV-CSE-2026-MP-14**
PROJECT GUIDE:	**Prof. Shweta Pandey**
EVALUATION SCOPE:	**UML Structural & Behavioral Views (30% Rubric Weightage)**
DELIVERABLES:	**Activity, Sequence, Class, Component, Deployment, Package, Object**
INTEGRITY SECURITY:	**SHA-256 Cryptographic Hash & Digital Signature Certified**

DEVELOPMENT TEAM MEMBERS (MP-14)

Mukul Bhamgare (23100BTCSAII14297) - Lead Architect / Backend & Scheduling Engine

Rashi Kashiv (23100BTCSAII14314) - Frontend Lead / UI Design & Matrix Component

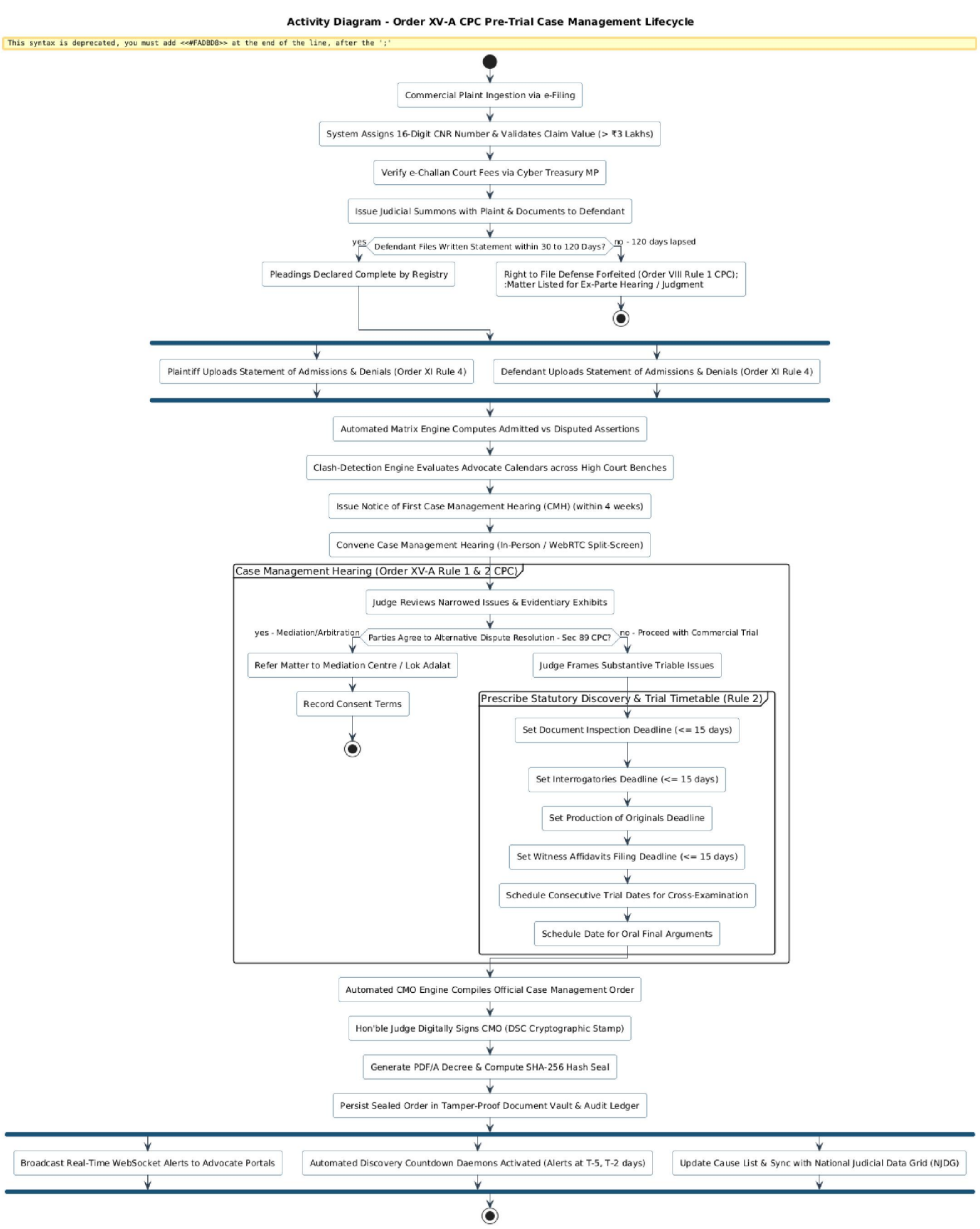
Sakshi Singh (23100BTCSAII14317) - Data Architect / Security & ORM Modeling

Kanishk (23100BTCSAII14293) - DevOps / QA, CI/CD Pipeline & CMO Engine

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1. UML Activity Diagram

Order XV-A CPC Pre-Trial Case Management Lifecycle & Decision Nodes

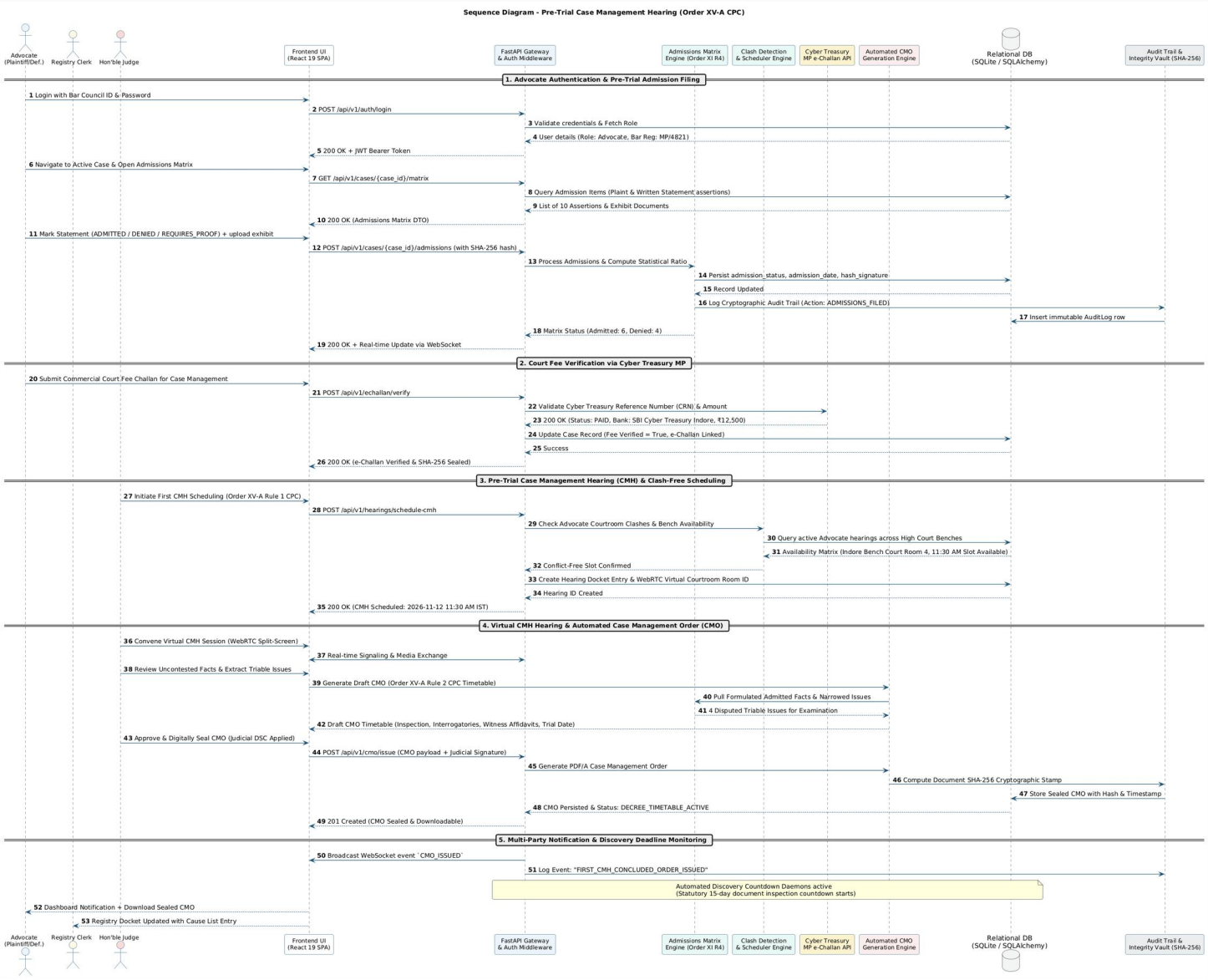


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Strict Statutory Limitation: Enforces mandatory 120-day maximum for Written Statements (Order VIII Rule 1 CPC).
- Order XI Rule 4 Matrix: Parallel itemized filing of statements of admissions and denials with documentary exhibits.
- Statutory Rule 2 Timetable: Discovery inspection (<=15d), interrogatories (<=15d), witness affidavits (<=15d).
- Continuous Cross-Examination: Fixes day-to-day continuous trial dates to eliminate non-statutory adjournments.

2. UML Sequence Diagram

Pre-Trial Case Management Hearing (CMH) & Multi-Party Interaction Flow

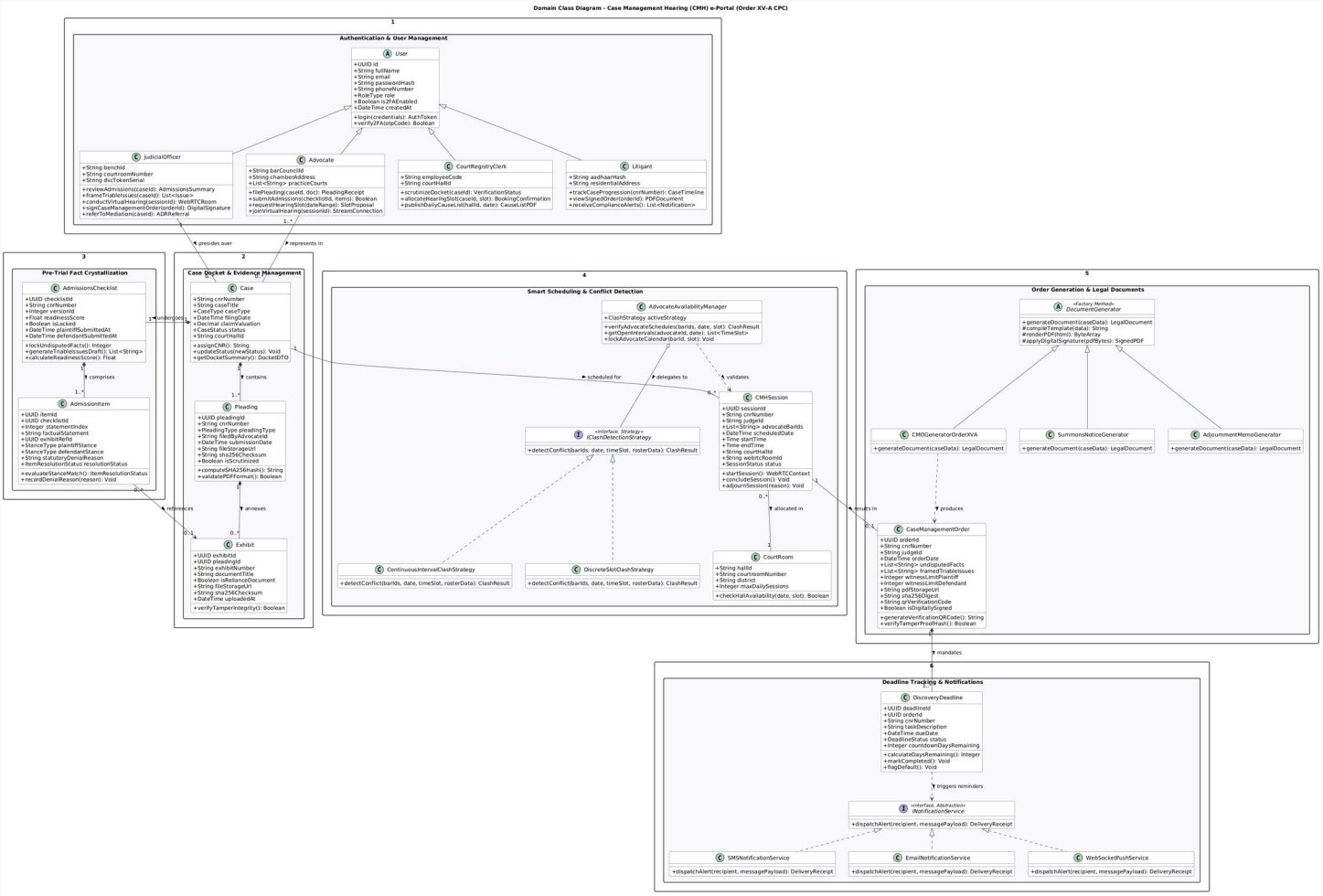


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Advocate Authentication: Bar Council credentials verified with RBAC JWT bearer tokens.
- Cyber Treasury MP: Real-time verification of Challan Reference Number (CRN) for commercial court fees.
- Clash-Free Scheduling: Cross-bench detection algorithms query advocate availability before slot reservation.
- Cryptographic Decree Sealing: Final Case Management Order stamped with SHA-256 seal and Judicial DSC.

3. UML Class Diagram

Domain Object-Oriented Architecture, Encapsulation, and ORM Entities

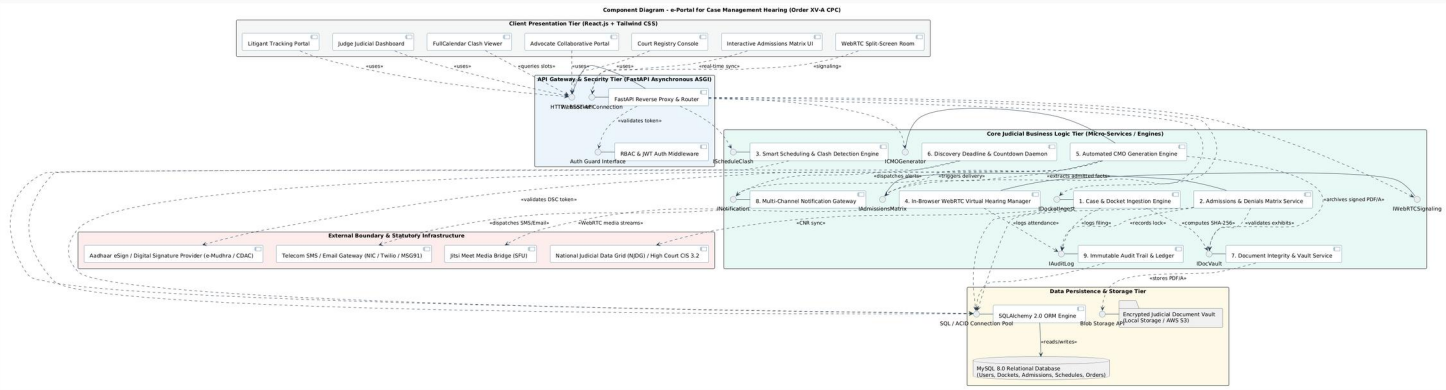


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Role Inheritance: Abstract User base class specialized into Judge, Advocate, Clerk, and Litigant domain models.
- Data Encapsulation: Strict private attributes for password hashes and DSC keys with public verification contracts.
- Statutory Associations: Case maintains 1:N AdmissionMatrix, 1:1 CaseManagementOrder, and 1:1 EChallanPayment.
- SQLAlchemy 2.0 ORM: Maps domain entities to relational tables with foreign keys and database constraints.

4. UML Component Diagram

5-Tier Decoupled Judicial Architecture & Micro-Engine Interface Contracts

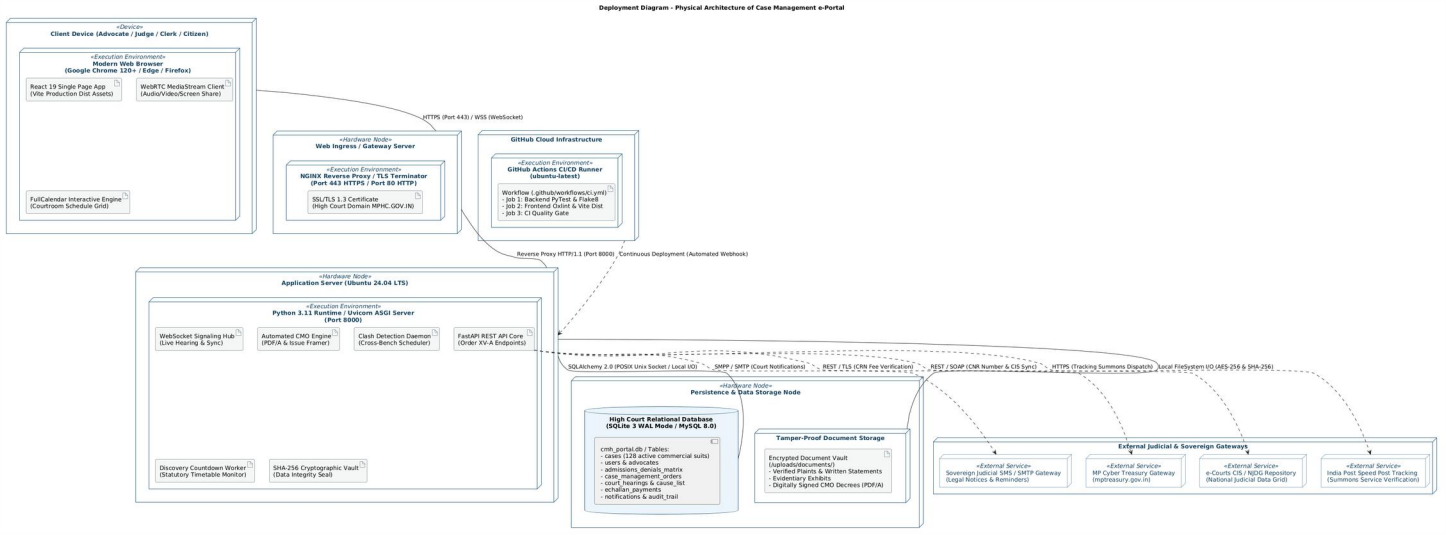


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Presentation Tier: React 19 Single Page App with Tailwind CSS, FullCalendar grid, and WebRTC courtroom.
- Security & Gateway Tier: Asynchronous FastAPI ASGI gateway with rate limiting and JWT auth middleware.
- Business Logic Tier: 9 decoupled services (Admissions Matrix, Clash Engine, CMO Builder, Countdown Daemon).
- Persistence & Gateways: SQLite WAL / MySQL database, encrypted document vault, Cyber Treasury, and NJDG.

5. UML Deployment Diagram

Hardware Node Topology, Execution Environments, and Secure Protocols

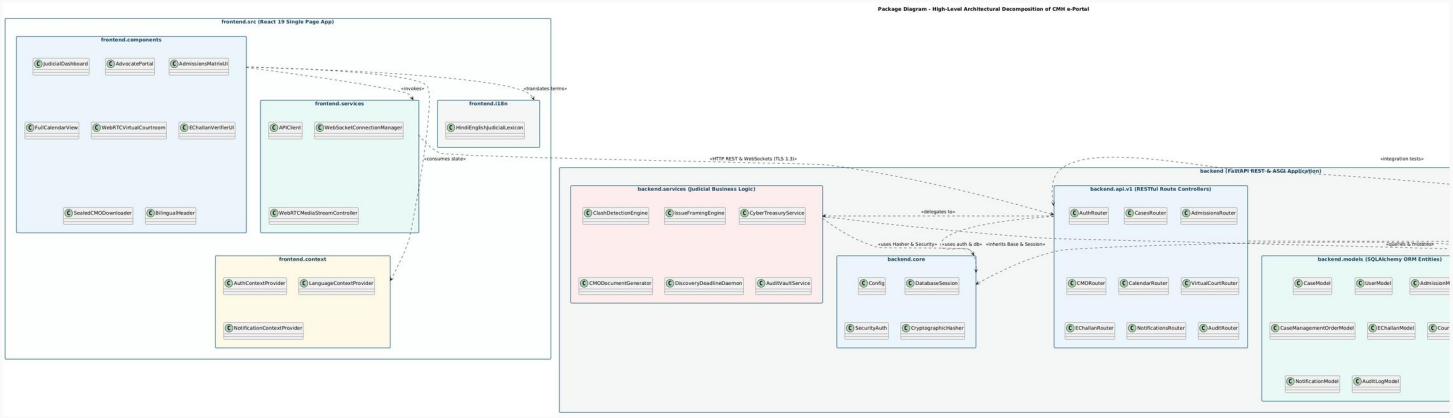


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Client Devices: Browser execution environment running React 19 SPA over TLS 1.3 HTTPS and WSS protocols.
- Web Ingress Node: NGINX reverse proxy providing SSL termination and protective DDoS rate limiting.
- Application Server Node: Ubuntu 24.04 LTS host running Python 3.11 with Uvicorn ASGI on port 8000.
- Storage & External Nodes: POSIX file vault for PDF/A exhibits, connecting to MP Treasury and India Post APIs.

6. UML Package Diagram

Modular Package Decomposition & Layered Architectural Separation

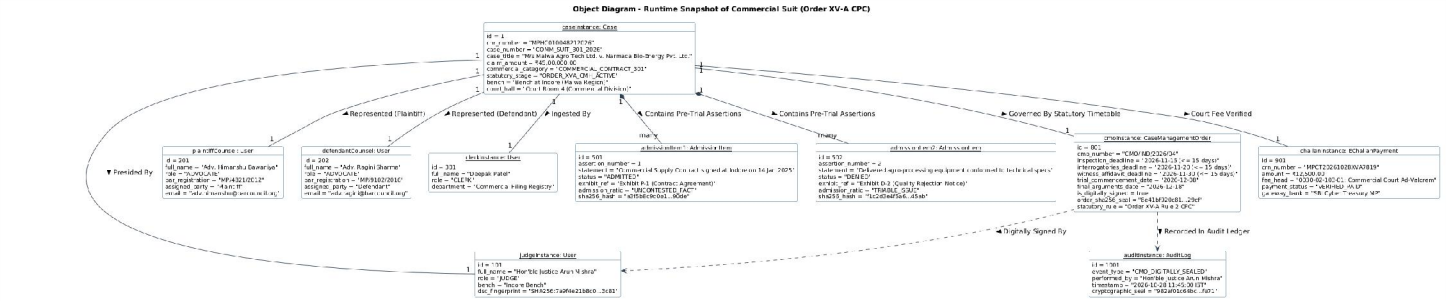


ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Layered Separation: Routers (api.v1) handle HTTP, Schemas validate DTOs, Services enforce business logic.
- Domain Models (models): Cleanly separated domain ORM entities with zero coupling to HTTP request handlers.
- Asynchronous Workers: Countdown daemon executes independently of the request-response cycle.
- Frontend Modularity: Client components cleanly decoupled into UI views, context providers, and API services.

7. UML Object Diagram

Runtime Instance Snapshot: Commercial Suit COMM_SUIT_301_2026 (Indore Bench)



ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Live Suit Instance: Case MPHC010048212026 (Malwa Agro Tech v. Narmada Bio-Energy, Specified Value: ₹45 Lakhs).
- Judicial Actors: Hon'ble Justice Arun Mishra (Judge), Adv. Himanshu Bawaniya (Plff), Adv. Ragini Sharma (Def).
- Admissions State: Commercial contract marked ADMITTED; technical specs conformity marked DENIED triable issue.
- Statutory Timetable: CMO/IND/2026/04 with mandatory discovery inspection deadline and SHA-256 seal.